Windmills: A Prerequisite Book

A prerequisite book for Windmills of the minds: an algorithm for Fermat's Two Squares Theorem

Hing Lun Chan

Assumed background: arithmetic (addition, subtraction, multiplication); high-school algebra; basic proof writing (induction, contradiction); sets and functions

Built with paper-to-book. Every definition and theorem is tagged in the margin with its source; quotations are checked automatically against the retrieved sources, and statements marked Lean ✓ are verified against Lean's mathlib.

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How to read this book

The chapters build, in order, everything the paper uses that goes beyond the assumed background. Each definition and theorem carries a margin note saying where it comes from:

- Source: `source`, §– quoted from a retrieved source, and the quotation was checked automatically;
- Lean ✓ `name` — the statement matches a declaration in Lean's mathlib that compiles;
- not yet formally checked: `name` — a formal counterpart exists but has not been checked yet;
- Derived here from sources — not stated in any source, but proved in this book from the sourced results named, with the proof given and checked in Lean;
- **UNSOURCED** or **LEDGER CHECK FAILED** (id) — not backed by a checked source: treat with suspicion.

Shaded boxes are definitions; passages with a grey bar on the left are verbatim quotations; blue boxes show where the idea appears in the paper. The last part is the paper itself.